

A question that evidence can answer

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Engineering demonstration and defense

Without opening the old lesson, explain one key idea from Engineering demonstration and defense. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for A question that evidence can answer. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for A question that evidence can answer. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

You have response times and correctness for two lookup procedures on ten fictional requests. Write a bounded question and one missing detail.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Proposal says prove AI is useful; available records are retrieval results for five room queries. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A measurable narrow question is not evidence of its answer. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

You have response times and correctness for two lookup procedures on ten fictional requests. Write a bounded question and one missing detail.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled a question that evidence can answer evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of a question that evidence can answer. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: A question that evidence can answer

A question with a measurable answer. A team wants to know whether giving a diagram changes accuracy on six fixed plant-identification tasks.

Candidate questions: "Are diagrams wonderful?" and "Does version B answer more of these six tasks correctly than version A?"

AI ACADEMY / AI-7/1

A question that evidence can answer

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Narrow the question

Name versions A and B, the six tasks and correct-answer scoring.



2. Define the comparison

Use the same tasks and labels with and without the diagram.



3. Bound the conclusion

The study can compare those versions on those tasks, not all diagrams or learners.

CHANGE ONE THING

Replace the six tasks with six easier tasks only for B.

Explain the changed result using the matching lesson.

Which proposed diagram question has a measurable outcome? Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A question with a measurable answer. A team wants to know whether giving a diagram changes accuracy on six fixed plant-identification tasks.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The study can compare those versions on those tasks, not all diagrams or learners.

1. Define the comparison
2. Bound the conclusion
3. Narrow the question

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Bound the conclusion The study can compare those versions on those tasks, not all diagrams or learners. An answerable research question connects a change to an observable outcome and a specified population. Opinion words need an operational definition before measurement.

Practice 2: explain the deciding evidence

Changed-case practice: Replace the six tasks with six easier tasks only for B. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. The comparison no longer isolates the diagram condition.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The comparison no longer isolates the diagram condition. Task difficulty now changes alongside the proposed intervention.

Repair a missing building block

Review Engineering demonstration and defense (ai-6/36). Goal: Explain the mechanism and terminology of engineering demonstration and defense. Short practice: Locate V7, room E and read-only scope in the demonstration. Which evidence supports the answer, and why must a request to email everyone still be refused?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
